

LOW STYRENE SBR PATENT LANDSCAPE: EMULSION, SOLUTION, AND FUNCTIONALIZED COPOLYMER ROUTES

1. ABSTRACT

This report synthesizes seven patents relevant to low styrene styrene-butadiene rubber (SBR), with emphasis on explicit low-styrene targets, styrene-content control, and process routes that preserve or improve performance in tire and rubber applications. The target is defined here as SBR with reduced styrene fraction, typically in the low single-digit to low-teens wt% range for emulsion SBR, or as low-to-moderate styrene solution SBR architectures designed to deliver low hysteresis, improved filler interaction, or tailored tread/bead-filler performance. The patent landscape spans two major synthesis families: emulsion polymerization of low-styrene SBR using emulsifier systems tuned to suppress coagulum and improve polymerization rate, and solution polymerization of styrene-butadiene copolymers using organolithium initiation, polar modifiers, and end-group functionalization or chain-end coupling to improve silica/carbon black interaction.

US10407566B2 is the clearest direct low-styrene disclosure, specifying 5-10 wt% styrene and an emulsifier mixture of aliphatic organic acid plus sulfonate-based compound, with Example 1 using 94:6 butadiene:styrene feed at 10°C and 80% conversion. EP2719549B1 and US7763696B2 are solution-polymerization disclosures that are not low-styrene-focused per se, but they provide important control over styrene incorporation, molecular weight, and microstructure through TMEDA/n-BuLi systems, continuous reactors, and salt-assisted cyclic-vinyl formation.

US6313232B1 shows continuous solution SBR with midstream and post-reactor tin addition to reduce Payne effect and hysteresis. US20110112212A1 is a tire composition patent using modified SBR and modified BR with silica/carbon black blends, relevant to low-styrene SBR because it demonstrates how modified SBR grades are deployed in tread formulations.

EP3732243A1 discloses low molecular weight tapered SBR with 20-30 wt% styrene and controlled styrene sequencing, while US8492479B2 teaches polyether-block functionalized diene copolymers, including SBR, for improved filler interaction and hysteresis reduction.

Overall, the evidence supports low-styrene SBR synthesis by emulsion polymerization with carefully balanced emulsifiers, or by solution polymerization with controlled monomer feed, polar modifier, and post-polymer functionalization to compensate for reduced styrene content in final tire performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10407566B2
- Patent Title: Method for preparing styrene-butadiene rubber having low combined styrene ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses emulsion polymerization of styrene and butadiene to prepare low styrene-butadiene rubber using a mixed emulsifier system composed of an aliphatic organic acid and a sulfonate-based compound. The process is explicitly directed to low combined styrene content SBR and includes latex coagulation, washing, dewatering, and drying to isolate the rubber. - Styrene content: 5% by weight to 10% by weight — Claim 2 - Sulfonate-based compound loading: 1 part by weight to 5 parts by weight per 100 parts by combined weight of styrene monomer and butadiene monomer — Claim 1 - Emulsifier weight ratio (aliphatic organic acid:sulfonate-based compound): 1:1 to 10:1 — Claim 1 - Aliphatic organic acid carbon chain length: 12 to 18 carbon atoms — Claim 3 - Example 1 styrene feed: 6 parts by weight — Example 1 - Example 1 butadiene feed: 94 parts by weight — Example 1 - Example 1 polymerization temperature: 10° C. — Example 1 - Example 1 initiator: 0.05 parts by weight — Example 1 - Example 1 molecular weight regulator: 0.5 parts by weight — Example 1 - Example 1 water charge: 200 parts by weight per 100 parts by weight monomer — Example 1 - Example 1 conversion ratio at termination: 80% — Example 1 - Example 1 calcium chloride addition: 2 parts by weight per 100 parts by weight solid latex — Example 1 - Example 1 coagulation aging temperature: 70° C. — Example 1 - Example 1 coagulation aging time: 20 minutes — Example 1 - Example 1 washing cycles: 2 to 3 times with ion exchange water — Example 1 - Example 1 polymerization time to 55% conversion: 7.1 hr — Table 1 - Comparative Example 2 polymerization time to 55% conversion: 14 hr — Table 1 - Example 3 produced coagulation: 0.005% by weight — Table 1

Relevant Experimental Evidence - Example 1: Low-styrene SBR latex was prepared at 10° C. from a 94:6 butadiene:styrene monomer feed and terminated at 80% conversion; the latex was coagulated with calcium chloride and dried to rubber. - Example 2: Reduced aliphatic organic acid and increased SANS loading were used, showing that emulsifier balance affects polymerization time and coagulum. - Example 3: Equal parts aliphatic organic acid and SANS gave the lowest produced coagulation in Table 1 (0.005% by weight). - Comparative Example 2: Using SANS

alone increased polymerization time to 14 hr while still giving low coagulation. - Comparative Example 1: Using aliphatic organic acid alone gave 7.5 hr polymerization time but much higher coagulation (0.3% by weight).

Technical Relevance This is the most directly relevant patent for low styrene SBR because it explicitly claims 5-10 wt% styrene and demonstrates how emulsifier selection controls polymerization rate and coagulum in low-styrene emulsion SBR. It provides concrete formulation and process guidance for making low-styrene rubber suitable for blending with high-styrene SBR in tire bead-filler applications.

Patent 2

- Patent Number: EP2719549B1
- Patent Title: Functionalized polymer, rubber composition and pneumatic tire
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of styrene and butadiene in a batch reactor followed by end-group functionalization using ethylene sulfide, chlorotriethoxysilane, or both in cascade. The resulting functionalized SBR is characterized for molecular weight, styrene/vinyl content, Tg, and compounding performance with silica and silane coupling agent. - Example 1 reactor volume: 1 gallon batch reactor — Example 1 - Example 1 polymerization temperature: 65 °C — Example 1 - Example 1 conversion threshold for termination: above 98% — Example 1 - Example 1 TMEDA:n-butyl lithium molar ratio: 4.9 — Example 1 - Example 1 n-BuLi in line inlet: 200 µmol per 100 g of monomers — Example 1 - Example 1 n-BuLi at reactor inlet: 700 µmol per 100 g of monomers — Example 1 - Example 1 overall Mn of functionalized SBR: 250,000 g/mol — Table 2 - Example 1 overall Mn of non-functionalized SBR: 181,000 g/mol — Table 2 - Example 1 overall Mn of ES functionalized SBR: 197,000 g/mol — Table 2 - Example 1 overall Mn of CTES functionalized SBR: 249,000 g/mol — Table 2 - Example 1 overall Mn of ES&CTES functionalized SBR: 206,000 g/mol — Table 2 - Example 1 Mooney viscosity of functionalized SBR: 65 — Table 3 - Example 1 Mooney viscosity of non-functionalized SBR: 35.9 — Table 3 - Example 1 Mooney viscosity of ES functionalized SBR: 32.4 — Table 3 - Example 1 Mooney viscosity of CTES functionalized SBR: 32.5 — Table 3 - Example 1 Mooney viscosity of ES and CTES functionalized SBR: 39.7 — Table 3 - Example 1 styrene content of functionalized SBR: 17.97 — Table 3 - Example 1 styrene content of non-functionalized SBR: 23.70 — Table 3 - Example 1 styrene content of ES functionalized SBR: 23.46 — Table 3 - Example 1 styrene content of CTES functionalized SBR: 23.73 — Table 3 - Example 1 styrene content of ES and CTES functionalized SBR: 25.44 — Table 3 - Example 1 Tg of functionalized SBR: -25.0 °C — Table 3 - Example 1 Tg of non-functionalized SBR: -22.08 °C — Table 3 - Example 1 Tg of ES functionalized SBR: -22.09 °C — Table 3 -

Example 1 Tg of CTES functionalized SBR: -22.00 °C — Table 3 - Example 1 Tg of ES and CTES functionalized SBR: -22.41 °C — Table 3 - Example 2 first non-productive mix stage: 4 minutes at 60 rpm — Example 2 - Example 2 first non-productive starting temperature: 140 °C — Example 2 - Example 2 second non-productive mix time: 3 minutes — Example 2 - Example 2 productive mix starting temperature: 60 °C — Example 2 - Example 2 productive mix time: 3 minutes — Example 2 - Example 2 silica loading: 65 phr — Example 2 - Example 2 oil loading: 30 phr — Example 2 - Example 2 stearic acid loading: 2 phr — Example 2 - Example 2 zinc oxide loading: 3.5 phr — Example 2 - Example 2 sulfur loading: 1.7 phr — Example 2 - Example 2 accelerators loading: 3.1 phr — Example 2 - Example 2 antidegradant loading: 0.75 phr — Example 2 - Example 2 silane coupling agent loading: 5.2 phr — Example 2 - Example 3 LAM/HAM ratio for control sample: 7.2 — Table 4 - Example 3 LAM/HAM ratio for non-functionalized control sample: 18.9 — Table 4 - Example 4 LAM/HAM ratio for control sample without silane: 17.5 — Table 5 - Example 5 S'max for comparative sample: 11.18 dNm — Table 6 - Example 5 S'max for control sample: 15.7 dNm — Table 6 - Example 6 S'max for comparative sample: 25.42 dN*m — Table 8

Relevant Experimental Evidence - Example 1: Batch solution polymerization at 65 °C with TMEDA and n-BuLi produced functionalized SBR samples with altered Mooney viscosity, styrene content, and Tg. - Example 2: Silica-filled compounds were mixed in a three-stage process, showing the formulation and processing route for evaluating functionalized SBR in tread-like compounds. - Example 3: With silane coupling agent, functionalized polymers showed reduced strain dependency relative to non-functionalized control in the Payne-effect analysis. - Example 4: Without silane coupling agent, functionalized polymers still altered filler interaction, but the control sample showed the lowest Payne effect. - Example 5: Productive compounds with silane showed higher torque response for functionalized polymers than the comparative sample. - Example 6: Productive compounds without silane showed that ES/CTES functionalization reduced Payne effect relative to non-functionalized polymer.

Technical Relevance Although not a low-styrene patent, it is relevant because it demonstrates solution-polymerized SBR with controlled styrene content and end-group functionalization to improve silica interaction and tire-relevant properties. Such functionalization strategies are useful when low styrene content is desired but filler reinforcement and hysteresis must still be maintained.

Patent 3

- Patent Number: US7763696B2
- Patent Title: Diene elastomer with cyclic vinyl units and processes for obtaining same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses continuous anionic solution polymerization of styrene-butadiene copolymers in stirred reactors using organolithium initiators, polar agents such as TMEDA, and in some embodiments alkali metal salts of aliphatic or alicyclic alcohols to increase cyclic vinyl units. The process is used to obtain linear or starred SBR with controlled styrene content, molecular weight, and microstructure. - Example 1 reactor volume: 0.8 litre perfectly stirred reactor — Example 1 - Example 1 solvent: cyclohexane — Example 1 - Example 1 monomer feed ratio (cyclohexane:butadiene:styrene:TMEDA): 100:12.2:2.1:0.057 by weight — Example 1 - Example 1 TMEDA concentration: approximately 500 ppm — Example 1 - Example 1 n-BuLi in line inlet: 200 μ mol per 100 g of monomers — Example 1 - Example 1 n-BuLi at reactor inlet: 700 μ mol per 100 g of monomers — Example 1 - Example 1 dwell time: 40 minutes — Example 1 - Example 1 temperature: 80 °C — Example 1 - Example 1 conversion rate: 98% — Example 1 - Example 1 inherent viscosity: 2.05 dl/g — Example 1 - Example 1 Mn: 130,000 g/mol — Example 1 - Example 1 Ip: 1.80 — Example 1 - Example 1 styrene linkages: 53%? No, 53% vinyl units in butadiene moiety and 10% vinylcyclopentane units — Example 1 - Example 1 butadiene moiety vinyl units: 53% — Example 1 - Example 1 butadiene moiety vinylcyclopentane units: 10% — Example 1 - Example 2 sodium tert.-amylate addition: sodium:active lithium ratio 0.2 — Example 2 - Example 2 TMEDA:n-BuLi ratio: 4.9 — Example 2 - Example 2 conversion rate: 97% — Example 2 - Example 2 inherent viscosity: 1.80 dl/g — Example 2 - Example 2 Mn: 120,000 g/mol — Example 2 - Example 2 Ip: 2.47 — Example 2 - Example 2 styrene linkages: 15% — Example 2 - Example 2 butadiene moiety vinyl linkages: 53% — Example 2 - Example 2 butadiene moiety vinylcyclopentane units: 23% — Example 2 - Example 3 reactor volume: 14 litre perfectly stirred reactor — Example 3 - Example 3 monomer feed ratio (cyclohexane:butadiene:styrene:TMEDA): 100:13.6:0.7:0.17 by weight — Example 3 - Example 3 TMEDA concentration: approximately 1500 ppm — Example 3 - Example 3 n-BuLi in line inlet: 300 μ mol per 100 g of monomers — Example 3 - Example 3 n-BuLi at reactor inlet: 500 μ mol per 100 g of monomers — Example 3 - Example 3 dwell time: 40 minutes — Example 3 - Example 3 temperature: 80 °C — Example 3 - Example 3 conversion rate: 84% — Example 3 - Example 3 inherent viscosity: 2.21 dl/g — Example 3 - Example 3 Mn of linear SBR: 135,000 g/mol — Example 3 - Example 3 Mn of starred SBR: 180,000 g/mol — Example 3 - Example 3 Ip: 1.85 — Example 3 - Example 3 styrene linkages: 5% — Example 3 - Example 3 butadiene moiety vinyl units: 60% — Example 3 - Example 3 butadiene moiety vinylcyclopentane units: 9% — Example 3 - Example 4 sodium tert.-amylate ratio: sodium:active lithium ratio 0.04 — Example 4 - Example 4 n-BuLi at reactor inlet: 660 μ mol per 100 g of monomers — Example 4 - Example 4 sodium:TMEDA molar ratio: 0.002 — Example 4 - Example 4 conversion rate: 90% — Example 4 - Example 4 inherent viscosity: 2.30 dl/g — Example 4 - Example 4 Mn of linear SBR: 150,000 g/mol — Example 4 - Example 4 TMEDA:n-BuLi ratio: 15.5 — Example 4 - Example 4 Mn of starred SBR: 260,000 g/mol — Example 4 - Example 4 Ip: 2.56 — Example 4 - Example 4 styrene linkages: 6% — Example 4 - Example 4 butadiene moiety vinyl linkages: 57% — Example 4 - Example 4 butadiene moiety vinylcyclopentane units: 22% — Example 4 - Example 5 cyclohexane charge: 14.3 parts per 100 parts of cyclohexane? as stated — Example 5 - Example 5 no styrene introduced — Example 5 - Example 5 n-BuLi at reactor inlet: 600 μ mol per 100 g of monomers — Example 5 - Example 5 TMEDA:n-BuLi ratio: 17.1 — Example 5 - Example 5 conversion rate: 92% — Example 5 - Example 5 inherent viscosity: 1.86 dl/g — Example 5 - Example 5 Mn of linear BR: 130,000 g/mol — Example 5 - Example 5 Mn of starred BR: 181,000 g/mol — Example 5 -

Example 5 Ip: 2.89 — Example 5 - Example 5 butadiene vinyl linkages: 56% — Example 5 - Example 5 butadiene vinylcyclopentane units: 20% — Example 5 - Example 6 monomer feed ratio (cyclohexane:butadiene:styrene:TMEDA): 100:13.15:1.13:0.29 by weight — Example 6 - Example 6 n-BuLi in line inlet: 200 μ mol per 100 g of monomers — Example 6 - Example 6 n-BuLi at reactor inlet: 1200 μ mol per 100 g of monomers — Example 6 - Example 6 TMEDA:n-BuLi ratio: 14.6 — Example 6 - Example 6 dwell time: 40 minutes — Example 6 - Example 6 temperature: 80 °C — Example 6 - Example 6 conversion rate: 92% — Example 6 - Example 6 inherent viscosity: 1.35 dl/g — Example 6 - Example 6 methyltrichlorosilane addition: 400 μ mol per 100 g of monomers — Example 6 - Example 6 Mn of linear SBR: 75,000 g/mol — Example 6 - Example 6 Mn of starred SBR: 120,000 g/mol — Example 6 - Example 6 Ip: 1.89 — Example 6 - Example 6 viscosity of starred polymer: 1.85 dl/g — Example 6 - Example 6 butadiene vinyl units: 51% — Example 6 - Example 6 butadiene vinylcyclopentane units: 18% — Example 6 - Example 7 monomer feed ratio (cyclohexane:butadiene:styrene:TMEDA): 100:12.30:2:0.35 by weight — Example 7 - Example 7 TMEDA concentration: approximately 3000 ppm — Example 7 - Example 7 n-BuLi in line inlet: 200 μ mol per 100 g of monomers — Example 7 - Example 7 n-BuLi at reactor inlet: 1500 μ mol per 100 g of monomers — Example 7 - Example 7 TMEDA:n-BuLi ratio: 14.1 — Example 7 - Example 7 dwell time: 40 minutes — Example 7 - Example 7 temperature: 80 °C — Example 7 - Example 7 conversion rate: 98% — Example 7 - Example 7 inherent viscosity: 0.90 dl/g — Example 7 - Example 7 Mn: 50,000 g/mol — Example 7 - Example 7 styrene linkages: 16% — Example 7 - Example 7 butadiene moiety vinyl units: 57% — Example 7 - Example 7 butadiene moiety vinylcyclopentane units: 17% — Example 7 - Example 8 sodium tert.-amylate ratio: sodium:active lithium ratio 0.4 — Example 8 - Example 8 sodium:TMEDA molar ratio: 0.05 — Example 8 - Example 8 conversion rate: 99% — Example 8 - Example 8 inherent viscosity: 0.43 dl/g — Example 8 - Example 8 TMEDA:n-BuLi ratio: 14.1 — Example 8 - Example 8 Mn: 6,000 g/mol — Example 8 - Example 8 styrene linkages: 16% — Example 8 - Example 8 butadiene moiety vinyl linkages: 43% — Example 8 - Example 8 butadiene moiety vinylcyclopentane units: 40% — Example 8

Relevant Experimental Evidence - Example 1: Continuous solution SBR without alkali metal salt gave 10% vinylcyclopentane units and Mn 130,000 g/mol. - Example 2: Adding sodium tert.-amylate increased vinylcyclopentane units to 23% and reduced inherent viscosity. - Example 4: At higher TMEDA:n-BuLi ratio and sodium tert.-amylate, starred SBR reached 22% vinylcyclopentane units and Mn 260,000 g/mol. - Example 6: Methyltrichlorosilane star-branching produced a starred SBR with 18% vinylcyclopentane units and Mn 120,000 g/mol. - Example 8: Low-Mn SBR with sodium tert.-amylate achieved 40% vinylcyclopentane units.

Technical Relevance This patent is relevant because it provides a continuous solution-polymerization route for styrene-butadiene copolymers with explicit control over styrene content and microstructure. While not a low-styrene patent in the strict sense, it offers process levers—TMEDA, n-BuLi, and optional alkali metal salt—that can be used to tune styrene incorporation and elastomer properties for low-styrene SBR design.

Patent 4

- Patent Number: US6313232B1
- Patent Title: Continuous addition of tin halides to solution SBR
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2001
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses continuous solution polymerization of butadiene/styrene in hydrocarbon solvent with organolithium initiation and staged addition of tin compounds during polymerization and again after withdrawal from the reactor. The process is intended to reduce Payne effect and hysteresis by modifying the polymer during and after polymerization. - Tin compound addition point: when degree of polymerization has reached between 25% and 80% — Claim 1 - First tin addition preferred point: between 50% and 80% polymerization — Claim 3 - Example sample A tin compound loading: 0.04 g active lithium per 100 g monomer equivalent — Comparative Example - Example sample B tin compound loading: 0.04 g active lithium per 100 g monomer equivalent — Comparative Example - Example bottle mixing time after tin addition: 15-30 minutes — Comparative Example - Example compounding temperature: 90 °C — Comparative Example - Example second mixing temperature: 105 °C — Comparative Example - Dynastat test temperature: 50 °C — Comparative Example - Strain sweep test temperature: 25 °C — Comparative Example - Strain sweep strain increment: 0.25% — Comparative Example - Strain sweep maximum strain: 14.7% — Comparative Example - Control sample Payne effect: $1.682 \text{ dyne/cm}^2 \times 10^{-7}$ — Comparative Example - Sample A Payne effect: $1.144 \text{ dyne/cm}^2 \times 10^{-7}$ — Comparative Example - Sample B Payne effect: $1.268 \text{ dyne/cm}^2 \times 10^{-7}$ — Comparative Example - Control Dynastat value: 0.168 — Comparative Example - Sample A Dynastat value: 0.150 — Comparative Example - Sample B Dynastat value: 0.147 — Comparative Example

Relevant Experimental Evidence - Comparative Example: Post-reactor tin treatment with Bu_3SnCl or SnCl_4 improved Dynastat and Payne-effect metrics versus control. - Comparative Example: Tin-treated samples were mixed with a natural-rubber/carbon-black compound and evaluated for hysteresis. - Comparative Example: Sample A showed 12.0% improvement in Dynastat and 32% improvement in $\Delta G'$ relative to control. - Comparative Example: Sample B showed 14.3% improvement in Dynastat and 25% improvement in $\Delta G'$ relative to control.

Technical Relevance This patent is relevant because it shows how solution SBR can be modified in a continuous process to lower hysteresis and Payne effect, which are key performance metrics when styrene content is reduced. The staged tin addition strategy is useful for tuning end-group chemistry and filler interaction in low-styrene SBR formulations.

Patent 5

- Patent Number: US20110112212A1

- Patent Title: Tire
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily a tire composition disclosure, but it includes production examples of modified conjugated diene polymers and modified conjugated diene-aromatic vinyl copolymers prepared by anionic polymerization and end-group modification with hydrocarbyloxysilane compounds. The resulting polymers are compounded with silica and carbon black to evaluate dry, wet, on-ice, and low-heat-generating performance. - Example 1 reactor pressure: around 2 bar nitrogen — Example 1 - Example 1 polymerization temperature: 40 °C — Example 1 - Example 1 conversion rate: 95% — Example 1 - Example 1 initial inherent viscosity: 1.01 dL/g — Example 1 - Example 1 Mn of control polymer: 82,000 g/mol — Example 1 - Example 1 final inherent viscosity: 1.75 dL/g — Example 1 - Example 1 Mn of block copolymer: 178,000 g/mol — Example 1 - Example 1 Ip: 1.39 — Example 1 - Example 2 reactor pressure: around 2 bar nitrogen — Example 2 - Example 2 polymerization temperature: 40 °C — Example 2 - Example 2 conversion rate: 96% — Example 2 - Example 2 initial inherent viscosity: 0.96 dL/g — Example 2 - Example 2 Mn of control polymer: 77,000 g/mol — Example 2 - Example 2 final inherent viscosity: 1.65 dL/g — Example 2 - Example 2 Mn of block copolymer: 167,000 g/mol — Example 2 - Example 2 Ip: 1.19 — Example 2 - Example 3 reactor pressure: around 2 bar nitrogen — Example 3 - Example 3 polymerization temperature: 40 °C — Example 3 - Example 3 conversion rate: 96% — Example 3 - Example 3 initial inherent viscosity: 1.00 dL/g — Example 3 - Example 3 Mn of control polymer: 84,000 g/mol — Example 3 - Example 3 final inherent viscosity: 1.61 dL/g — Example 3 - Example 3 Mn of block copolymer: 151,000 g/mol — Example 3 - Example 3 Ip: 1.23 — Example 3 - Example 4 reactor pressure: around 2 bar nitrogen — Example 4 - Example 4 polymerization temperature: 45 °C — Example 4 - Example 4 conversion rate: 92% — Example 4 - Example 4 ML viscosity: 58 — Example 4 - Example 4 Mn of copolymer: 161,000 g/mol — Example 4 - Example 4 Ip: 1.09 — Example 4 - Example 4 styrene content: 25% — Example 4 - Example 4 trans-1,4 units: 20% — Example 4 - Example 4 cis-1,4 units: 19% — Example 4 - Example 4 1,2-units: 61% — Example 4 - Example 4 (CH₃)₂Si functional groups: 3.73 mmol/kg — Example 4 - Example 5 reactor pressure: around 2 bar nitrogen — Example 5 - Example 5 polymerization temperature: 45 °C — Example 5 - Example 5 conversion rate: 92% — Example 5 - Example 5 ML viscosity: 54 — Example 5 - Example 5 Mn of copolymer: 158,000 g/mol — Example 5 - Example 5 Ip: 1.09 — Example 5 - Example 5 styrene content: 26% — Example 5 - Example 5 trans-1,4 units: 20% — Example 5 - Example 5 cis-1,4 units: 19% — Example 5 - Example 5 1,2-units: 61% — Example 5

Relevant Experimental Evidence - Example 1: Polyether-block SBR with two SiOH functional groups was prepared and coupled, increasing inherent viscosity from 1.01 to 1.75 dL/g. - Example 2: Pendant polyether-block SBR with one SiOH group was prepared and coupled, giving Mn

167,000 g/mol. - Example 3: Pendant polyether-block SBR without SiOH was prepared, showing a lower final viscosity increase than Example 2. - Example 4: A control SBR with SiOH at the chain end was prepared at 45 °C and characterized by 25% styrene content. - Example 5: Non-functional SBR control was prepared under the same reactor conditions as Example 4.

Technical Relevance This patent is relevant because it demonstrates how solution-polymerized SBR can be functionalized to improve filler interaction and hysteresis, which is important when designing low-styrene grades that otherwise lose reinforcement. The disclosed styrene contents and polymerization conditions provide a useful benchmark for low-to-moderate styrene SBR architectures.

Patent 6

- Patent Number: EP3732243A1
- Patent Title: Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low molecular weight tapered styrene-butadiene copolymers prepared by solution polymerization of styrene and butadiene in hydrocarbon solvent using an organolithium initiator and sodium dodecylbenzenesulfonate. The copolymers are designed as partial or full oil replacements in tire rubber compositions to improve storage modulus and wear resistance. - Claim 1 total styrene content: 20 to 30 percent by weight — Claim 1 - Claim 1 total butadiene content: 70 to 80 percent by weight — Claim 1 - Claim 1 styrene sequences >3 units: greater than 30 weight percent of total styrene content — Claim 1 - Claim 1 styrene sequences >10 units: less than 20 weight percent of total styrene content — Claim 1 - Claim 5 number average molecular weight: 30,000 to 80,000 — Claim 5 - Claim 6 glass transition temperature: -80°C to -70°C — Claim 6 - Rubber composition reinforcing filler: 30 to 100 phr — Description - Rubber composition process oil: up to 10 phr — Description - Rubber composition tapered SBR loading: 1 to 15 phr — Description - Example 1 hexane charge: 0.18 kg — Example 1 - Example 1 styrene charge: 0.36 kg of 31.8 wt% in hexane — Example 1 - Example 1 butadiene charge: 1.68 kg of 20.2 wt% in hexane — Example 1 - Example 1 SDBS charge: 2.27 mL of 0.5 M in ethylbenzene — Example 1 - Example 1 n-butyl lithium charge: 6.05 mL of 1.60 M in hexane — Example 1 - Example 1 reactor jacket temperature: 60°C — Example 1 - Example 1 batch temperature peak: 69°C — Example 1 - Example 1 additional reaction time: 2 hours — Example 1 - Example 1 Mn: 52 kg/mol — Example 1 - Example 1 Mw: 55 kg/mol — Example 1 - Example 1 PDI: 1.05 — Example 1 - Example 1 Tg: -76.07°C — Example 1 - Example 1 percent styrene: 26.6% — Example 1 - Example 1 vinyl content: 9.4% — Example 1 - Example 2 hexane charge: 0.30 kg — Example 2 - Example 2 styrene charge: 0.55 kg of 32.9 wt% in hexane — Example 2 - Example 2 butadiene charge: 2.72

kg of 20 wt% in hexane — Example 2 - Example 2 SDBS charge: 3.63 mL of 0.5 M in ethylbenzene — Example 2 - Example 2 n-butyl lithium charge: 9.07 mL of 1.60 M in hexane — Example 2 - Example 2 reactor jacket temperature: 66°C — Example 2 - Example 2 batch temperature peak: 107°C — Example 2 - Example 2 additional reaction time: 40 minutes — Example 2 - Example 2 Mn: 66 kg/mol — Example 2 - Example 2 Mw: 74 kg/mol — Example 2 - Example 2 PDI: 1.12 — Example 2 - Example 2 Tg: -78.79°C — Example 2 - Example 2 percent styrene: 25.7% — Example 2 - Example 2 vinyl content: 9.7% — Example 2

Relevant Experimental Evidence - Example 1: Tapered SBR with 26.6% styrene and Mn 52 kg/mol improved wear resistance and storage modulus when used as an oil replacement. - Example 2: Similar tapered SBR with 25.7% styrene and Mn 66 kg/mol improved wear resistance in both carbon-black and silica-filled compounds. - Claim set: The copolymer is defined by 20-30 wt% styrene and 70-80 wt% butadiene with controlled styrene sequencing. - Description: The copolymer is intended for tire rubber compositions with 30-100 phr filler and up to 10 phr process oil.

Technical Relevance This patent is relevant because it defines a styrene-butadiene copolymer with controlled styrene distribution and relatively low molecular weight, which is useful for low-styrene SBR-like performance tuning. Although the styrene level is higher than classic low-styrene emulsion SBR, the sequencing and solution-polymerization approach are directly informative for balancing stiffness, wear, and processability.

Patent 7

- Patent Number: US8492479B2
- Patent Title: Process for preparing a diene copolymer comprising a polyether block, diene ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of diene block copolymers containing at least one polyether block linked through silicon chemistry to a living diene elastomer, including SBR. The process uses anionic polymerization of styrene and butadiene in hydrocarbon solvent, followed by reaction with functional polyethers and subsequent coupling/hydrolysis steps to create filler-interactive block copolymers. - Example 1 polyethylene glycol monomethyl ether Mn: approximately 516 g/mol — Example 1 - Example 1 KOH: 5.61 g, 0.1 mol — Example 1 - Example 1 allyl bromide: 8.7 mL, 0.1 mol — Example 1 - Example 1 hexamethylcyclotrisiloxane solution concentration: 0.2 mol·L⁻¹ — Example 1 - Example 1 poly(oxy-1,2-ethanediyl) functional agent concentration: 0.22 mol·L⁻¹ — Example 1 - Example 1 reactor volume: 10-liter reactor — Example 1 - Example 1 nitrogen pressure: around 2 bar — Example 1 - Example 1 methylcyclohexane charge: 4200 g — Example 1 - Example 1 styrene

charge: 156 g — Example 1 - Example 1 butadiene charge: 444 g — Example 1 - Example 1 tetrahydrofurfuryl ether charge: 0.45 mL — Example 1 - Example 1 n-BuLi charge: 5.1 mmol — Example 1 - Example 1 polymerization temperature: 40 °C — Example 1 - Example 1 conversion time: 74 min — Example 1 - Example 1 conversion rate: 95% — Example 1 - Example 1 initial inherent viscosity: 1.01 dL/g — Example 1 - Example 1 Mn of control: 82,000 g/mol — Example 1 - Example 1 Ip of control: 1.10 — Example 1 - Example 1 hexamethylcyclotrisiloxane addition: 8.4 mL of 0.2 mol·L⁻¹ solution — Example 1 - Example 1 second coupling reagent addition: 11.8 mL of 0.22 mol·L⁻¹ solution — Example 1 - Example 1 coupling reaction time: 90 min — Example 1 - Example 1 coupling reaction temperature: 60 °C — Example 1 - Example 1 final inherent viscosity: 1.75 dL/g — Example 1 - Example 1 viscosity jump: 1.73 — Example 1 - Example 1 ML viscosity: 55 — Example 1 - Example 1 Mn of block copolymer: 178,000 g/mol — Example 1 - Example 1 Ip of block copolymer: 1.39 — Example 1 - Example 1 antioxidant loading: 0.8 phr and 0.2 phr — Example 1 - Example 2 reactor pressure: around 2 bar — Example 2 - Example 2 methylcyclohexane charge: 5500 g — Example 2 - Example 2 styrene charge: 143 g — Example 2 - Example 2 butadiene charge: 407 g — Example 2 - Example 2 tetrahydrofurfuryl ether charge: 0.57 mL — Example 2 - Example 2 n-BuLi charge: 5.1 mmol — Example 2 - Example 2 polymerization temperature: 40 °C — Example 2 - Example 2 conversion time: 71 min — Example 2 - Example 2 conversion rate: 96% — Example 2 - Example 2 initial inherent viscosity: 0.96 dL/g — Example 2 - Example 2 Mn of control: 77,000 g/mol — Example 2 - Example 2 Ip of control: 1.06 — Example 2 - Example 2 coupling reagent concentration: 0.14 mol·L⁻¹ — Example 2 - Example 2 coupling reaction temperature: -15 °C — Example 2 - Example 2 coupling reaction time: 90 min — Example 2 - Example 2 final inherent viscosity: 1.65 dL/g — Example 2 - Example 2 viscosity jump: 1.72 — Example 2 - Example 2 ML viscosity: 54 — Example 2 - Example 2 Mn of block copolymer: 167,000 g/mol — Example 2 - Example 2 Ip of block copolymer: 1.19 — Example 2 - Example 3 coupling reagent concentration: 0.20 mol·L⁻¹ — Example 3 - Example 3 methylcyclohexane charge: 4200 g — Example 3 - Example 3 styrene charge: 156 g — Example 3 - Example 3 butadiene charge: 444 g — Example 3 - Example 3 tetrahydrofurfuryl ether charge: 0.45 mL — Example 3 - Example 3 n-BuLi charge: 4.5 mmol — Example 3 - Example 3 polymerization temperature: 40 °C — Example 3 - Example 3 conversion time: 69 min — Example 3 - Example 3 conversion rate: 96% — Example 3 - Example 3 initial inherent viscosity: 1.00 dL/g — Example 3 - Example 3 Mn of control: 84,000 g/mol — Example 3 - Example 3 coupling reaction time: 60 min — Example 3 - Example 3 coupling reaction temperature: 60 °C — Example 3 - Example 3 final inherent viscosity: 1.61 dL/g — Example 3 - Example 3 viscosity jump: 1.61 — Example 3 - Example 3 ML viscosity: 50 — Example 3 - Example 3 Mn of block copolymer: 151,000 g/mol — Example 3 - Example 3 Ip of block copolymer: 1.23 — Example 3 - Example 4 reactor pressure: around 2 bar — Example 4 - Example 4 methylcyclohexane charge: 48.9 kg — Example 4 - Example 4 styrene charge: 1.93 kg — Example 4 - Example 4 butadiene charge: 5.21 kg — Example 4 - Example 4 tetrahydrofurfuryl ether solution concentration: 0.065 mol·L⁻¹ — Example 4 - Example 4 n-BuLi solution concentration: 0.065 mol·L⁻¹ — Example 4 - Example 4 polymerization temperature: 45 °C — Example 4 - Example 4 conversion time: 45 min — Example 4 - Example 4 conversion rate: 92% — Example 4 - Example 4 hexamethylcyclotrisiloxane solution concentration: 0.0253 mol·L⁻¹ — Example 4 - Example 4 coupling reaction temperature: 60 °C — Example 4 - Example 4 coupling reaction time: 30 min — Example 4 - Example 4 ML viscosity: 58 — Example 4 - Example 4 Mn of copolymer: 161,000 g/mol — Example 4 - Example

4 Ip: 1.09 — Example 4 - Example 4 trans-1,4 units: 20% — Example 4 - Example 4 cis-1,4 units: 19% — Example 4 - Example 4 1,2-units: 61% — Example 4 - Example 4 styrene content: 25% — Example 4 - Example 4 (CH₃)₂Si functional groups: 3.73 mmol/kg — Example 4 - Example 5 reactor pressure: around 2 bar — Example 5 - Example 5 methylcyclohexane charge: 48.9 kg — Example 5 - Example 5 styrene charge: 1.93 kg — Example 5 - Example 5 butadiene charge: 5.21 kg — Example 5 - Example 5 tetrahydrofurfuryl ether solution concentration: 0.065 mol·L⁻¹ — Example 5 - Example 5 n-BuLi solution concentration: 0.065 mol·L⁻¹ — Example 5 - Example 5 polymerization temperature: 45 °C — Example 5 - Example 5 conversion time: 45 min — Example 5 - Example 5 conversion rate: 92% — Example 5 - Example 5 methanol solution concentration: 0.15 mol·L⁻¹ — Example 5 - Example 5 ML viscosity: 54 — Example 5 - Example 5 Mn of copolymer: 158,000 g/mol — Example 5 - Example 5 Ip: 1.09 — Example 5 - Example 5 trans-1,4 units: 20% — Example 5 - Example 5 cis-1,4 units: 19% — Example 5 - Example 5 1,2-units: 61% — Example 5 - Example 5 styrene content: 26% — Example 5

Relevant Experimental Evidence - Example 1: A polyether-block SBR with two SiOH groups was prepared, coupled, and shown to increase inherent viscosity and molecular weight. - Example 2: A pendant polyether-block SBR with one SiOH group was prepared and coupled at -15 °C, giving a high Mn block copolymer. - Example 3: A pendant polyether-block SBR without SiOH was prepared as a control for coupling behavior. - Example 4: A control SBR with SiOH chain-end functionality was prepared at 45 °C and characterized by 25% styrene content. - Example 5: A non-functional SBR control was prepared under the same conditions as Example 4.

Technical Relevance This patent is relevant because it shows how solution-polymerized SBR can be functionalized with polyether blocks to improve interaction with reinforcing fillers and reduce hysteresis. Such functionality is especially useful for low-styrene SBR formulations, where filler coupling and tread performance must compensate for reduced styrene-derived stiffness.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US10407566B2 is the only patent in the set that explicitly targets low styrene SBR at 5-10 wt% styrene; the others mainly use moderate styrene levels or focus on functionalization rather than low styrene content.
- US10407566B2 uses emulsion polymerization with an aliphatic organic acid/sulfonate emulsifier pair, whereas EP2719549B1, US7763696B2, US6313232B1, EP3732243A1, and US8492479B2 rely on solution polymerization or anionic living-polymer routes.
- US10407566B2 emphasizes latex coagulation and washing after emulsion polymerization; the solution patents emphasize reactor residence time, organolithium initiation, polar modifiers, and post-polymer end-group modification or coupling.
- US7763696B2 and EP2719549B1 both use organolithium/TMEDA-type solution polymerization, but US7763696B2 uniquely adds alkali metal salts to raise cyclic vinyl content, while EP2719549B1 focuses on ES/CTES end-group functionalization.

- US6313232B1 differs from the others by using staged tin addition during and after continuous polymerization to reduce Payne effect and hysteresis rather than directly controlling styrene content.
- EP3732243A1 specifies a tapered SBR with 20-30 wt% styrene and low molecular weight, which is higher in styrene than the low-styrene target but relevant for sequencing control and oil-replacement behavior.
- US8492479B2 and EP2719549B1 both aim to improve filler interaction through functionalization; US8492479B2 uses polyether blocks linked via silicon chemistry, while EP2719549B1 uses ES/CTES chain-end modification.
- Across the solution patents, temperatures cluster around 40-80 °C, with continuous or batch reactors, organolithium initiators, and hydrocarbon solvents being common; emulsion low-styrene SBR is instead run at low temperature with emulsifier optimization.

4. CONCLUSION

The evidence supports two practical synthesis directions for low styrene SBR. The most direct route is emulsion polymerization as disclosed in US10407566B2, where 5-10 wt% styrene, a 94:6 butadiene:styrene feed in Example 1, and a mixed aliphatic organic acid/sulfonate emulsifier system gave low coagulum and acceptable polymerization time. For solution-based routes, the patents show that low-styrene or styrene-controlled SBR can be made by organolithium polymerization with polar modifiers, continuous reactors, and post-polymer functionalization to recover filler interaction and hysteresis performance. Recommended evidence-based parameters are: styrene content near 5-10 wt% for true low-styrene emulsion SBR; low-temperature emulsion polymerization around 10 °C with emulsifier blending; or, for solution SBR, 40-80 °C polymerization with organolithium initiation, TMEDA or similar polar agents, and optional end-group functionalization or tin/silicon modification to improve tire-relevant properties.

5. REFERENCES

- US10407566B2 | Method for preparing styrene-butadiene rubber having low combined styrene ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10407566B2/en>
- EP2719549B1 | Functionalized polymer, rubber composition and pneumatic tire | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2719549B1/en>
- US7763696B2 | Diene elastomer with cyclic vinyl units and processes for obtaining same | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7763696B2/en>
- US6313232B1 | Continuous addition of tin halides to solution SBR | Not disclosed | US | 2001 | <https://patents.google.com/patent/US6313232B1/en>
- US20110112212A1 | Tire | Not disclosed | US | 2011 | <https://patents.google.com/patent/US20110112212A1/en>
- EP3732243A1 | Synthesis of low molecular weight tapered styrene-butadiene copolymer and its ... | Not disclosed | EP | 2020 | <https://patents.google.com/patent/EP3732243A1/en>

- US8492479B2 | Process for preparing a diene copolymer comprising a polyether block, diene ... | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8492479B2/en>